SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5-Coding Challenge**	**Assessment:**	**Assessment Tool 2– Coding Challenge**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	T.Megha Harsha	**Reg. No.:**	192424389
Section / Batch:	AI&DS	**Date:**	26-08-2026

Code of Conduct

I T.Megha Harsha (192424389) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: MEGHA

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

1. Implement diagnostic checks for the assumptions of linear regression from scratch: (a) linearity via residual-vs-fitted plots, (b) homoscedasticity via a formal test (e.g., Breusch-Pagan, coded manually), (c) normality of residuals via Q-Q plot and Shapiro-Wilk statistic, and (d) independence via Durbin-Watson statistic. Summarize which assumptions are violated for a given dataset and why.

Aim:

Implement diagnostic checks for Linear Regression assumptions and identify whether any assumptions are violated.

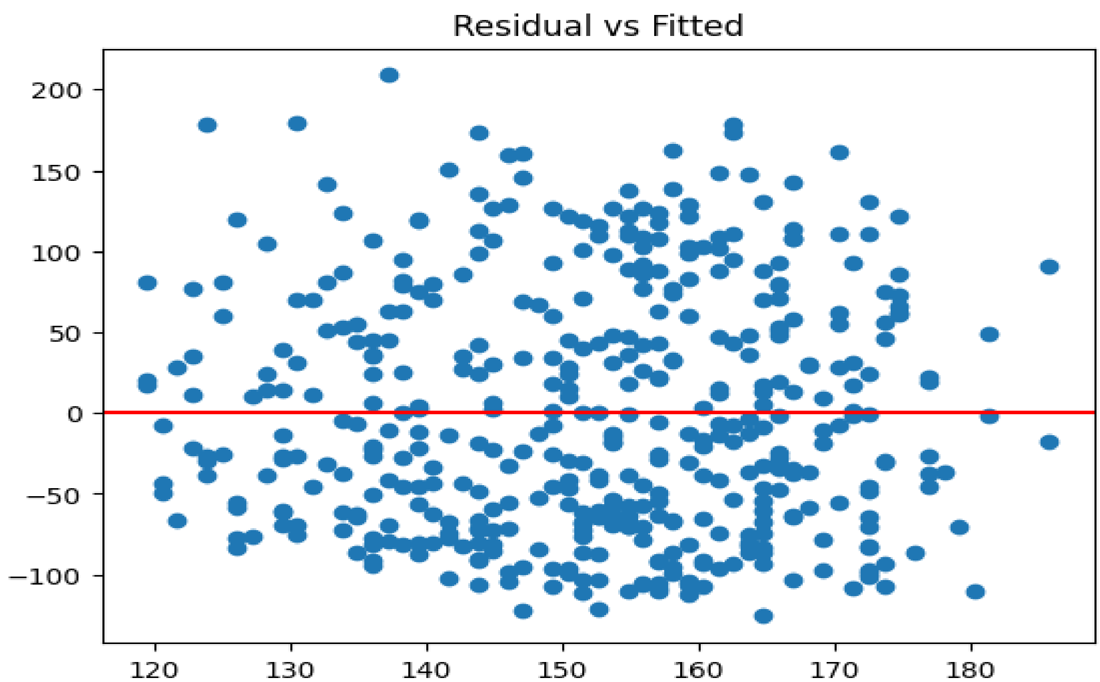
Code:

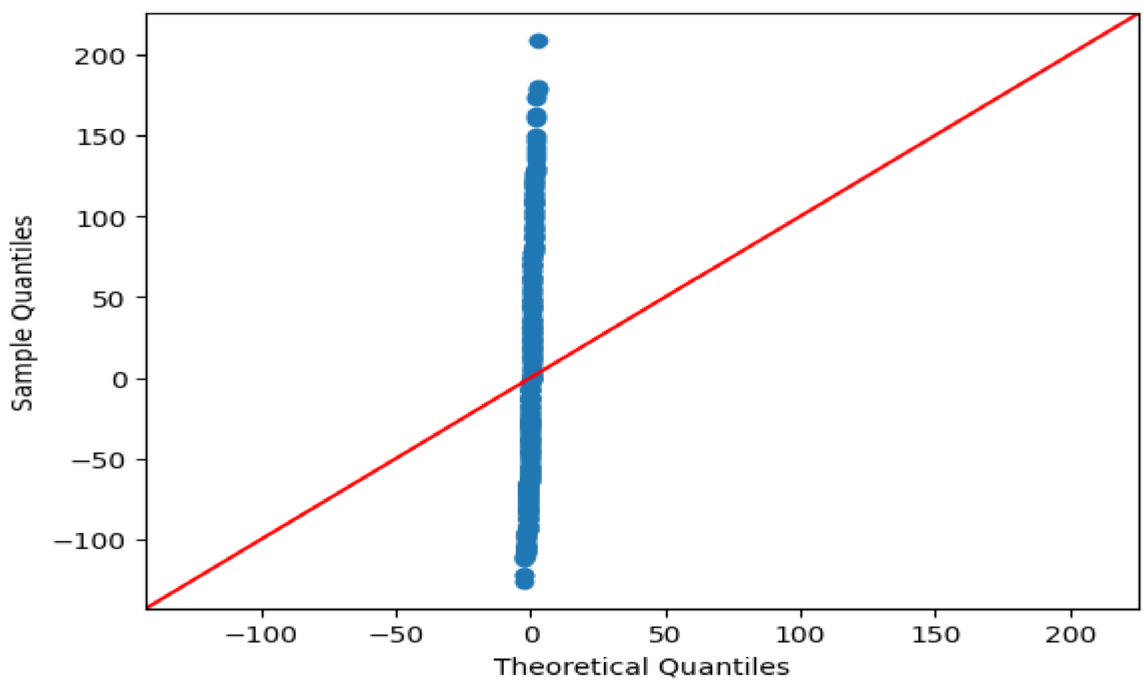
```python
import matplotlib.pyplot as plt
import statsmodels.api as sm
from scipy.stats import shapiro
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.stattools import durbin_watson
from sklearn.datasets import load_diabetes
# Load Dataset
data = load_diabetes()
X = data.data[:, 0:1]
y = data.target
# Linear Regression
X1 = sm.add_constant(X)
model = sm.OLS(y, X1).fit()
fitted = model.fittedvalues
residuals = model.resid
# Residual vs Fitted Plot
plt.scatter(fitted, residuals)
plt.axhline(y=0, color='r')
plt.title("Residual vs Fitted")
plt.show()
# Breusch-Pagan Test
bp = het_breuschpagan(residuals, X1)
# Q-Q Plot
sm.qqplot(residuals, line='45')
plt.show()
# Shapiro-Wilk Test
shapiro_test = shapiro(residuals)
# Durbin-Watson Test
dw = durbin_watson(residuals)
print("Breusch-Pagan p-value:", bp[1])
print("Shapiro-Wilk p-value:", shapiro_test.pvalue)
print("Durbin-Watson Statistic:", dw)
if bp[1] > 0.05:
    print("Homoscedasticity satisfied")
else:
    print("Homoscedasticity violated")

if shapiro_test.pvalue > 0.05:
    print("Residuals are normal")
else:
    print("Residuals are not normal")

if 1.5 < dw < 2.5:
    print("Residuals are independent")
else:
    print("Autocorrelation exists")
```

Output:






```
Breusch-Pagan p-value: 0.4147848464206191
Shapiro-Wilk p-value: 1.6297811895859195e-10
Durbin-Watson Statistic: 1.9212492321153118
Homoscedasticity satisfied
Residuals are not normal
Residuals are independent
```

Geotag Photo: